

Improvements in hostility and depression in relation to dietary change and cholesterol lowering. The Family Heart Study.

OBJECTIVE: To describe changes in negative emotions among participants of a cholesterol-lowering study. **DESIGN:** Cohort study. Quantitative evaluation of changes in negative emotions in relation to diet and plasma cholesterol levels before and after a 5-year dietary intervention program aimed at reducing plasma cholesterol levels. **SETTING:** Community-dwelling families of the Family Heart Study, Portland, Oregon. **PARTICIPANTS:** One hundred forty-nine men and 156 women from 233 families (mean age, 37.7 years). **MEASUREMENTS:** Changes in negative emotions including depression and aggressive hostility as measured by the Hopkins Symptom Checklist (SCL-90). **RESULTS:** Improvement in overall emotional state was noted for the entire sample. Those who consumed a low-fat, high complex-carbohydrate diet at the end of the study showed significantly greater improvements in depression ($P = 0.044$; difference in improvement, 2.9 points) and aggressive hostility ($P = 0.035$; difference in improvement, 3.3 points) as well as a reduction in their plasma cholesterol levels ($P = 0.024$; difference in improvement, 2.7%) compared with those who ate a high-fat "American diet." **CONCLUSIONS:** Participation in a cholesterol-lowering program may not be associated with a worsening in emotional state. To the contrary, improvements in diet appear to be associated with reductions in depression and aggressive hostility as well as with lowered plasma cholesterol levels.